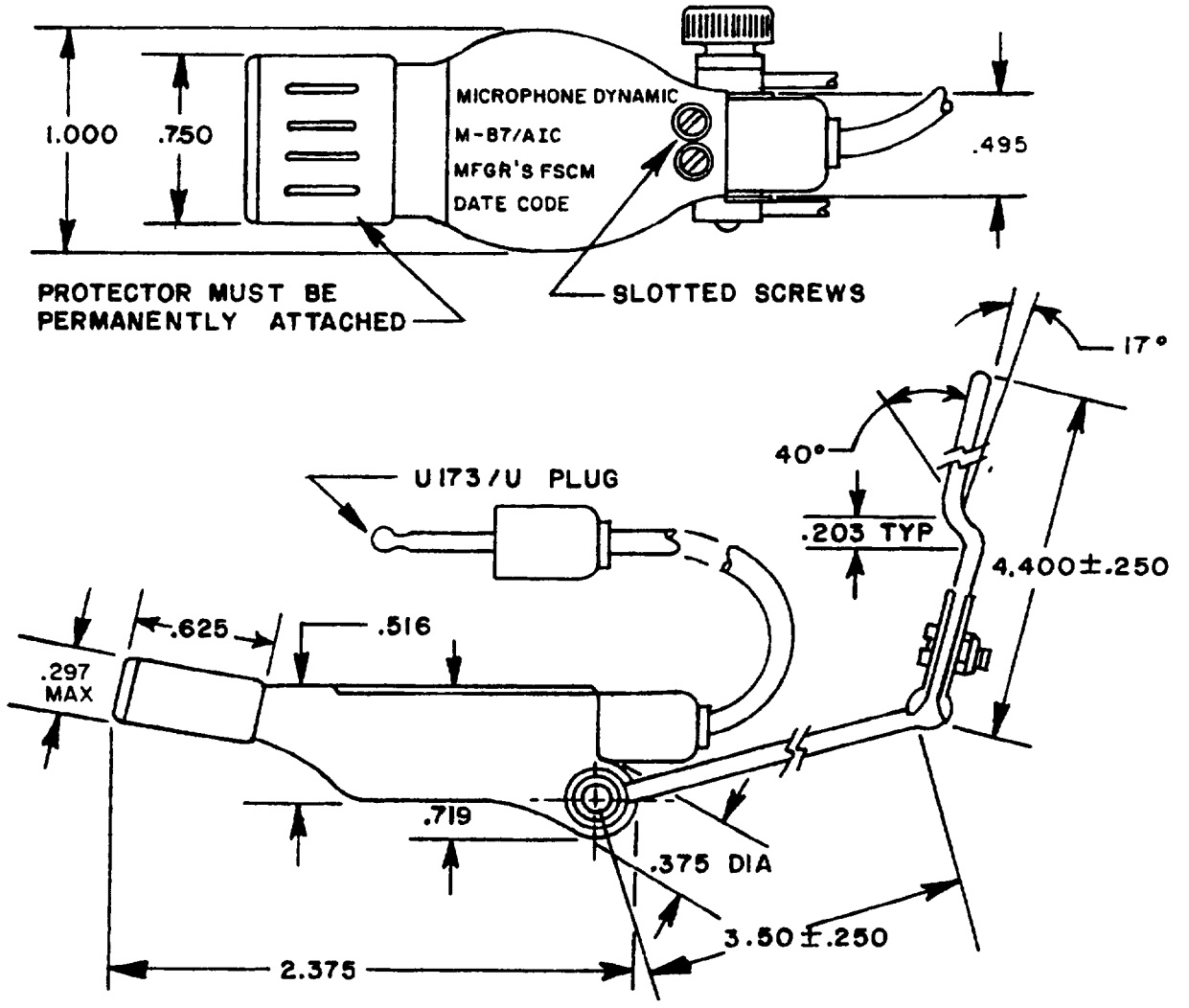
MILITARY SPECIFICATION SHEET

MICROPHONES, M87/AIC, M26542/2-01, M26542/2-02, AND M26542/2-03

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the microphone described herein shall consist of this specification and the latest issue of MIL-M-26542.

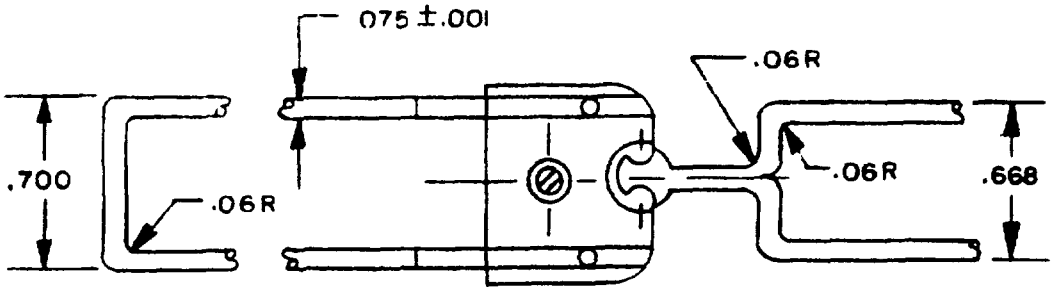


© FIGURE 1. Microphone assembly.

© denotes changes

FSC 5965

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TOP VIEW OF BALL AND SOCKET JOINT

INCHES	MM	INCHES	MM
.001	0.03	.516	13.11
.015	0.38	.625	15.88
06	1.5	.668	16.97
.0?5	1.91	.700	17.78
.?3	?.16	.719	18.26
.250	6 35	.750	19.05
297	7.54	1.000	25.40
75	9.52	2.375	60.32
.	12.57	3.500	88.90
		4.400	111.76

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.015 (0.38 mm).

(C) · URE 1 Microphone assembly - Continued.

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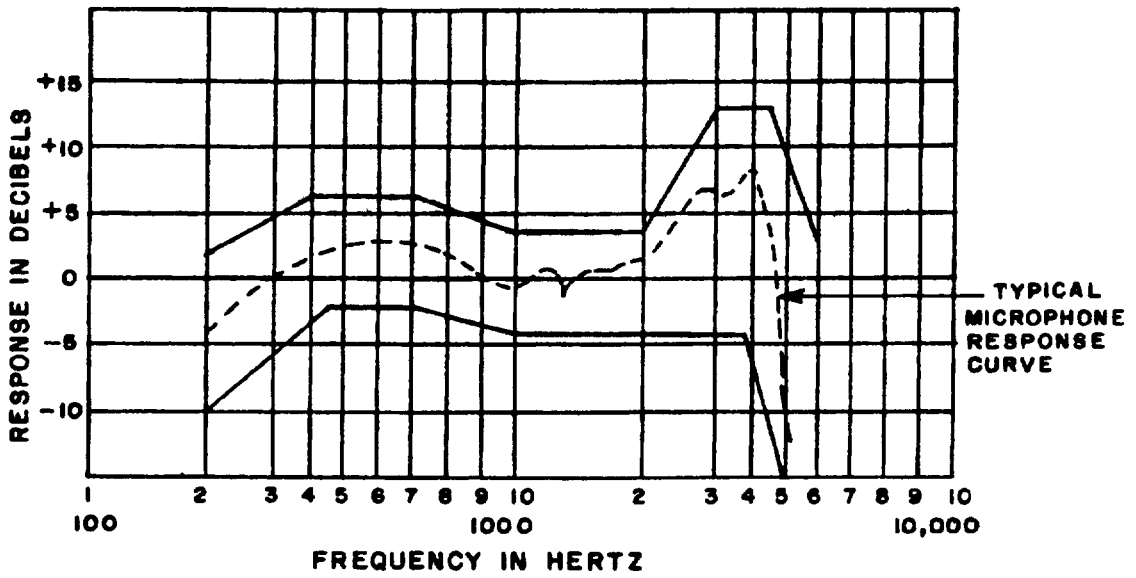


FIGURE 2. Frequency response at sea level.

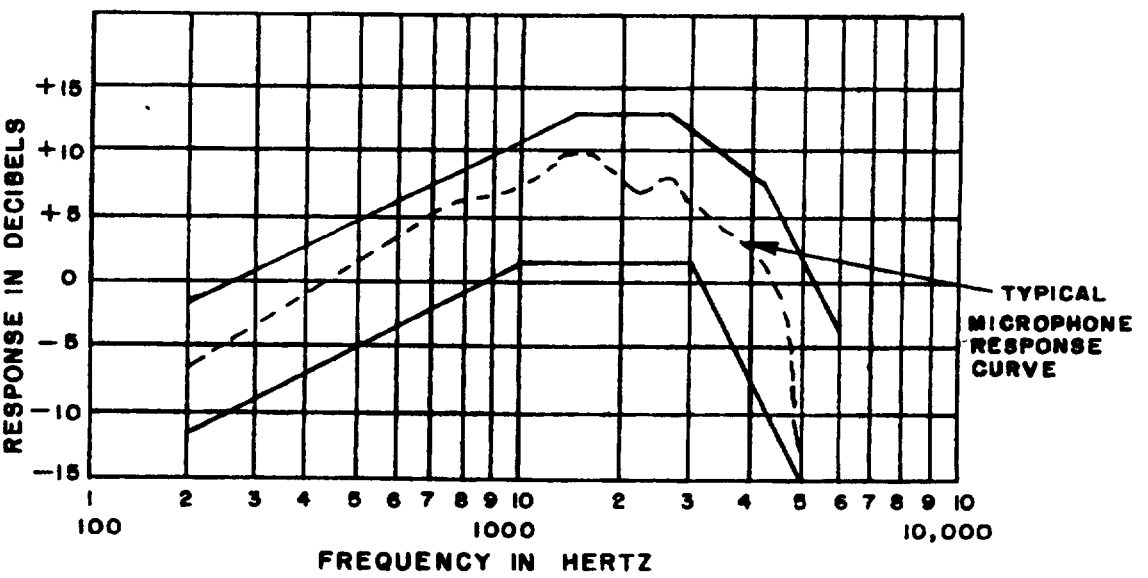


FIGURE 3. Frequency response at 25,000 feet.

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REQUIREMENTS:

Color: Semigloss black, 27038 in accordance with FED-STD-595.

Component parts:

Microphone element: See figure 1.

Boom: See figure 1.

© Cord assembly:

For M-87/AIC use cord assembly M22442/36-1 (6.25").

For M26542/2-01 use cord assembly M22442/36-3 (13").

For M26542/2-02 use cord assembly M22442/36-4 (16.25").

For M26542/2-03 use cord assembly M22442/36-6 (23").

Plug assembly. U-173/U.

Weight: Not to exceed 45 grams including boom but excluding cord assembly.

Performance:

Sensitivity at ground level: 1.8 to 4.0 microvolts referred to 74 dB relative to 20 micropascals with the sound port 0.18 inch (4.7 mm) from and coaxial with the opening of the artificial voice.

Sensitivity at altitude: Sensitivity shall be within 3 dB relative to the ground level sensitivity when tested at 25,000 ft.

Frequency response at ground level: See figure 2.

Frequency response at altitude: See figure 3.

Impedance: 4-6 ohms.

Resistive load: 5 ohms.

Intended use: Microphone, dynamic M-87/AIC is a noise-canceling dynamic microphone designed for use on a headband type headset at low altitudes or for use in a pressure type oxygen helmet, at altitudes where the use of an oxygen helmet is required. The microphone is intended for use as a part of intercommunication set AN/AIC-10 ( ) to provide communication under the noise conditions encountered in military aircraft.

The microphone assembly shall be tested in accordance with the tests listed in table I.

Marking: Microphone assemblies shall be marked as shown on figure 1 and shall additionally be identified with a heat sealed band on the cord assembly showing the words "microphone assembly" and the microphone assembly part number.

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© TABLE I. Parameter applicability.

Inspection	Requirements paragraph	Test paragraph	Qualification tests	Group "A" tests	Group "B" tests	Group "C" tests
Group I						
Visual and mechanical inspection	3.1,3.4,3.5	4.7.1	X	X		
Sensitivity of ground level	3.5.1	4.7.2.2	X	X		
Sensitivity at altitude	3.5.2	4.7.2.3	X			
Frequency response at ground level	3.5.3	4.7.2.4	X	X		
Frequency response at altitude	3.5.4	4.7.2.5	X			
Impedance	3.5.5	4.7.3	X	X		
Noise cancellation characteristics	3.5.7.1	3.7.5.1	X			
Effect of external magnetic field	3.5.8	4.7.6	X			
Stray magnetic field	3.5.9	4.7.7	X			
Linearity	3.5.10	4.7.8	X			
Talk out	3.5.11	4.7.9	X	X		
Dielectric withstanding voltage	3.5.12	4.7.10	X			
Signal to noise	3.5.7	4.7.5	X		X	
Distortion	3.5.6	4.7.4	X		X	
Interchangeability	3.5.13	4.7.11	X		X	
Group II						
Thermal shock	3.5.14	4.7.12	X			X
Humidity	3.5.15	4.7.13	X			X
Drop	3.5.16	4.7.14	X			X
Pressure equalization	3.5.17	4.7.15	X			X
Explosive decompression	3.5.18	4.7.16	X			X
Salt fog	3.5.19	4.7.17	X			X
Group III						
Vibration	3.5.20	4.7.18	X			X
Bounce	3.5.21	4.7.19	X			X
Altitude	3.5.22	4.7.20	X			X
Moisture barrier	3.5.23	4.7.21	X			X
Immersion	3.5.24	4.7.22	NA			
Group IV						
Fungus	3.5.25	4.7.23	NA			
Group V						
Gun blast	3.5.26	4.7.24	NA			

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Navy - AS
Air Force - 99
DLA - ES

User activities:
Army - AR, AT, AV, ME
Navy - OS, AS

Agent:
DLA - ES

Preparing activity:
Air Force - 85

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